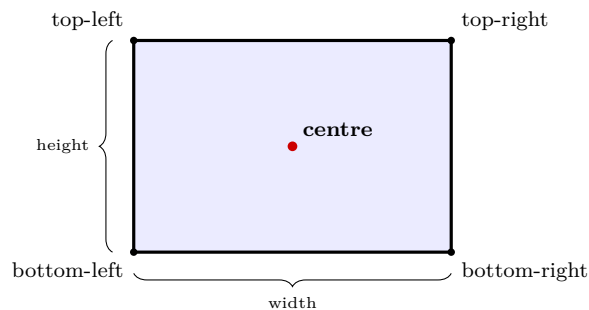


Answer key

A point is a spot. A sprite is a box. A point has two numbers: its x (how far across) and its y (how far up), written (x, y) . But a sprite in Scratch is not a single spot. It is a *box* that covers a patch of the grid. So how does Scratch know where the box sits? It pins the box to the grid by *one* point: the **centre**. When Scratch says a sprite is at (x, y) , it means the *centre* of the box is at (x, y) .

The box has other important points too. It has four **corners** and four **edges**. Each one has its own coordinates.



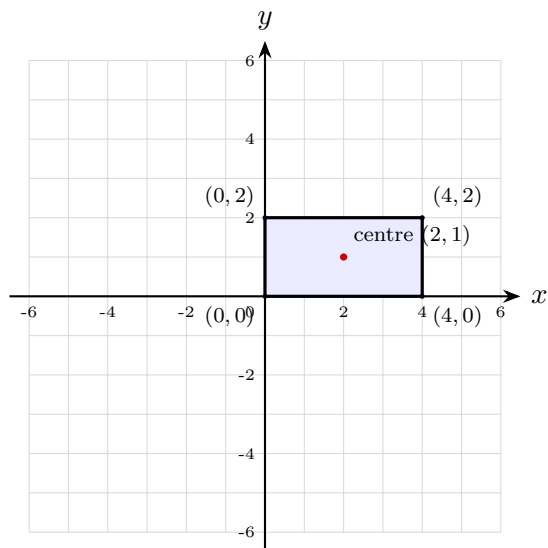
Finding a corner from the centre. A corner is half a width across and half a height up or down from the centre. So:

$$\text{corner } x = \text{centre } x \pm (\text{width} \div 2) \qquad \text{corner } y = \text{centre } y \pm (\text{height} \div 2)$$

Why this matters for the Dino game. To check if the Dino is standing on the ground, you must look at its *feet*: the bottom edge. You cannot look at the centre, because the centre sits higher up. The feet are at $(\text{centre } y) - (\text{height} \div 2)$. Scratch only tells you the centre with **y position**, so you subtract half the height yourself. Get this wrong and the Dino floats above the floor or sinks into it.

Problem 1.

Read the box. The box below has its centre marked. It is **4** wide and **2** tall. Write the coordinates of each part.

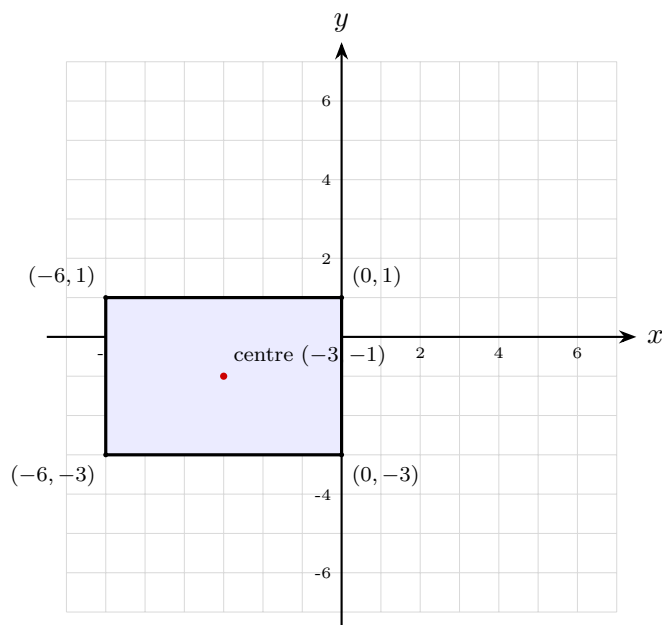


the bottom-left corner	(0, 0)
the bottom-right corner	(4, 0)
the top-left corner	(0, 2)
the top-right corner	(4, 2)

Problem 2.

Read the box. The box below has its centre marked. It is **6** wide and **4** tall. Write the coordinates of each part.

Two of these are corners (two numbers each). Two are whole edges, so they need only one number.

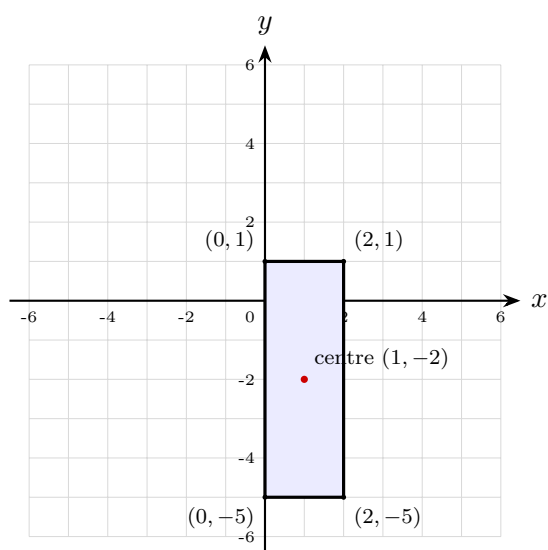


the top-left corner	$(-6, 1)$
the bottom-right corner	$(0, -3)$
the bottom edge (the feet)	$y = -3$
the top edge	$y = 1$

Problem 3.

Read the box. The box below has its centre marked. It is **2** wide and **6** tall. Write the coordinates of each part.

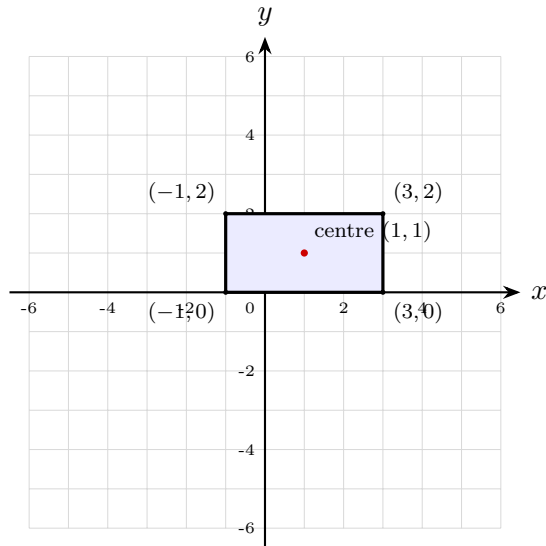
A tall, thin box. The left edge and the right edge each sit on one up-down line, so each has a single x .



the bottom-left corner	$(0, -5)$
the top-right corner	$(2, 1)$
the left edge	$x = 0$
the right edge	$x = 2$

Problem 4.

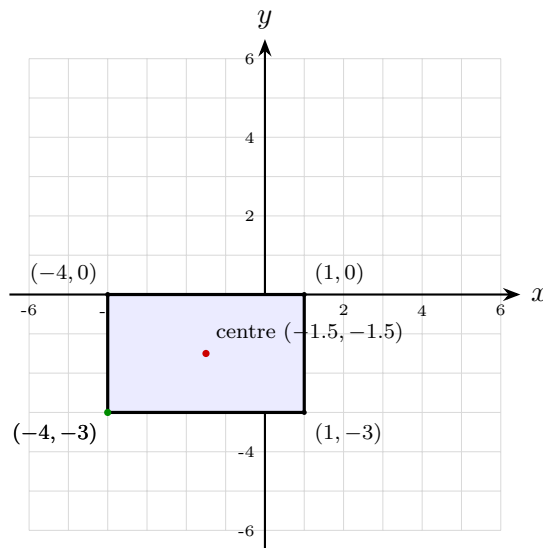
Place the box. Draw a box that is 4 wide and 2 tall, with its **centre** at $(1, 1)$. Mark the four corners.



Problem 5.

Place the box. This time you are given a *corner*, not the centre. Draw a box 5 wide and 3 tall whose **bottom-left corner** is at $(-4, -3)$.

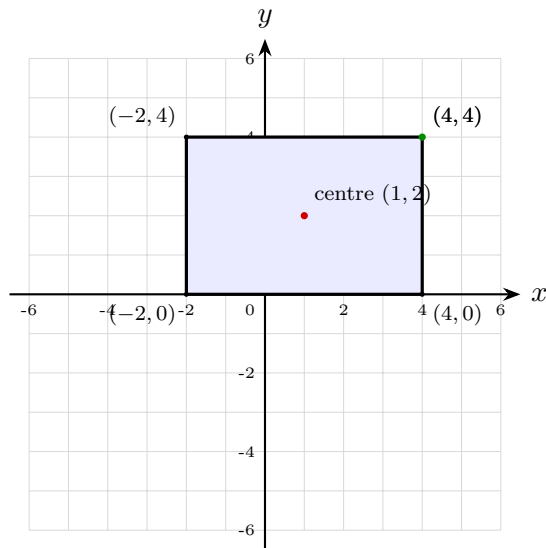
Same kind of numbers as before, but the box lands in a different place. From the bottom-left corner the box goes to the right and up.



Notice the centre is not at $(-4, -3)$ here. Choosing a different anchor point moves the whole box.

Problem 6.

Place the box. Draw a box 6 wide and 4 tall whose **top-right corner** is at $(4, 4)$. From the top-right corner the box goes to the left and down.



Problem 7.

Work it out. A box has its centre at $(0, 0)$. It is **4** wide and **2** tall. There is no picture this time. Work out the coordinates using the rule above.

Half the width is $4 \div 2 = 2$; half the height is $2 \div 2 = 1$.

the top-right corner	$(2, 1)$
the bottom-left corner	$(-2, -1)$
the bottom edge (the feet)	$y = -1$

Problem 8.

Work it out. A box has its centre at $(5, 3)$. It is **2** wide and **4** tall. There is no picture this time. Work out the coordinates using the rule above.

the bottom-right corner	$(6, 1)$
the top-left corner	$(4, 5)$
the top edge	$y = 5$

Problem 9.

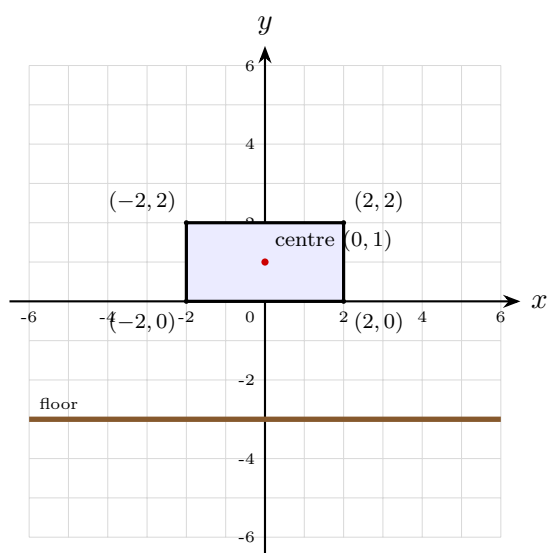
Work it out. A box has its centre at $(-2, 4)$. It is **8** wide and **6** tall. There is no picture this time. Work out the coordinates using the rule above.

A big box. Half the width is 4, half the height is 3.

the bottom-left corner	$(-6, 1)$
the bottom-right corner	$(2, 1)$
the top-left corner	$(-6, 7)$
the top-right corner	$(2, 7)$
the bottom edge (the feet)	$y = 1$

Problem 10.

Above or below? The brown line is the floor. Is the box above it, resting on it, or below it? Look at the feet (the bottom edge), not the centre.

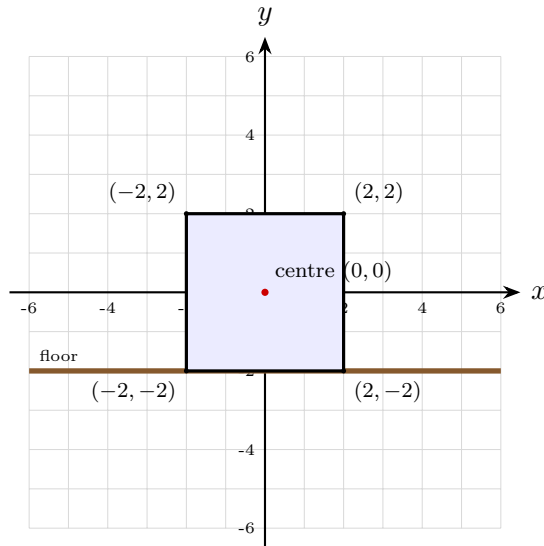


Answer: the box is **above the floor**.

The feet are at centre y minus height $\div 2 = 1 - 1 = 0$. The floor is at -3 . Since 0 is greater than -3 , the box is above the floor.

Problem 11.

Above or below? Is this box above the floor, on it, or below it? Work out where the feet are first.



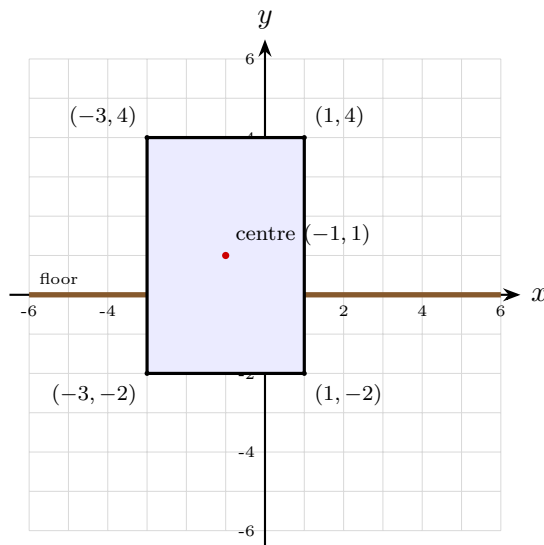
Answer: the box is **resting on the floor**.

The feet are at centre y minus height $\div 2 = 0 - 2 = -2$. The floor is at -2 . Since -2 is equal to -2 , the box is resting on the floor.

Problem 12.

Above or below? And this tall box? Its centre looks like it is well above the floor. Check the feet before you decide.

The centre can be above the floor while the feet are already below it. That is exactly the trap the Dino game sets.

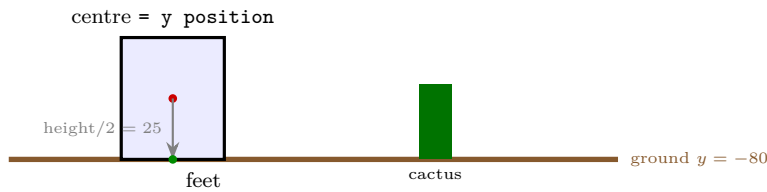


Answer: the box is **below the floor (sunk in)**.

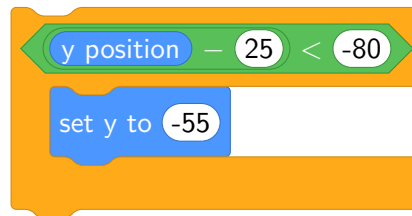
The feet are at centre y minus height $\div 2 = 1 - 3 = -2$. The floor is at 0 . Since -2 is less than 0 , the box is below the floor (sunk in).

Problem 13.

Write the block. Now the real Scratch stage. The floor is at $y = -80$. The Dino is 50 tall, so y position minus 25 is its feet. Write the **if** that is true when the Dino has landed on the floor. When it lands, set its centre to -55 so the feet rest on -80 .



The block:

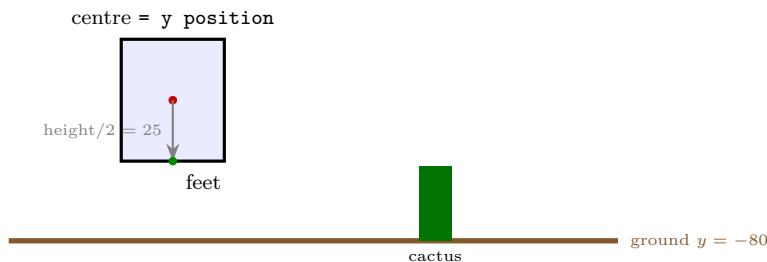


The feet are y position -25 . They reach the floor when that value drops to -80 , so the test is $(y \text{ position}) - 25 < -80$. Setting the centre to -55 puts the feet exactly on -80 ($-55 - 25 = -80$).

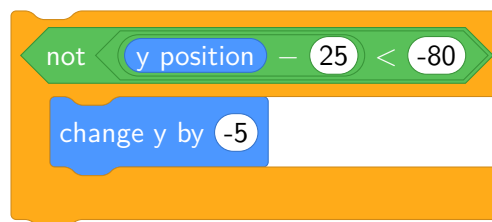
Problem 14.

Write the block. While the Dino is jumping it is *above* the floor. Write the **if** that is true while the Dino is still in the air (its feet have *not* reached the floor yet). While it is, pull it down by 5.

*"Still in the air" is the opposite of "landed". You can wrap the landed test in **not**.*



The block:

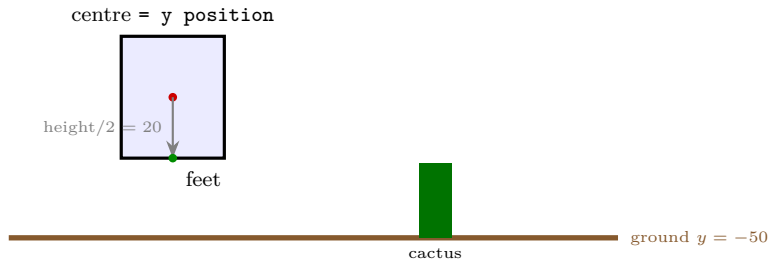


*Landed is $(y \text{ position}) - 25 < -80$. Still in the air is the opposite, so wrap it in **not**. Pulling down by 5 each time is gravity.*

Problem 15.

Write the block. A different check. The top of the cactus is at $y = -50$, and this Dino is 40 tall (so subtract 20 for its feet). Write the **if** that is true when the Dino's feet drop below the top of the cactus. If they do, broadcast **game over**.

Same shape as before, but a different line (-50 , not -80) and a different height (subtract 20, not 25).



The block:



The feet are $y \text{ position} - 20$. Below the cactus top means less than -50 : $(y \text{ position}) - 20 < -50$. You wrote the same pattern three times with different numbers. That is the whole grounded-check idea.
